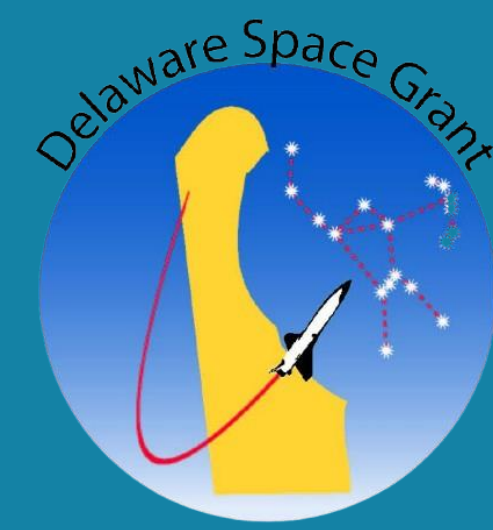




Pixels vs. Semantics: Does AI Cluster Emojis Like Humans Do?



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Abstract

Emojis are visual, but they carry deep semantic meaning. So how does AI see them? This study investigates the alignment between artificial intelligence and human perception in emoji categorization. Using 3,793 Apple emojis, we compared vision-based (**EfficientNetB3**) and text-based (**SentenceTransformer**) unsupervised clustering models against human card-sort data to determine which modality better captures human perceptual grouping. Results indicate that text-based clustering more closely aligned with human categorization, revealing that meaning-based features better reflect the way humans naturally organize visual information than visual features alone.

Methodology

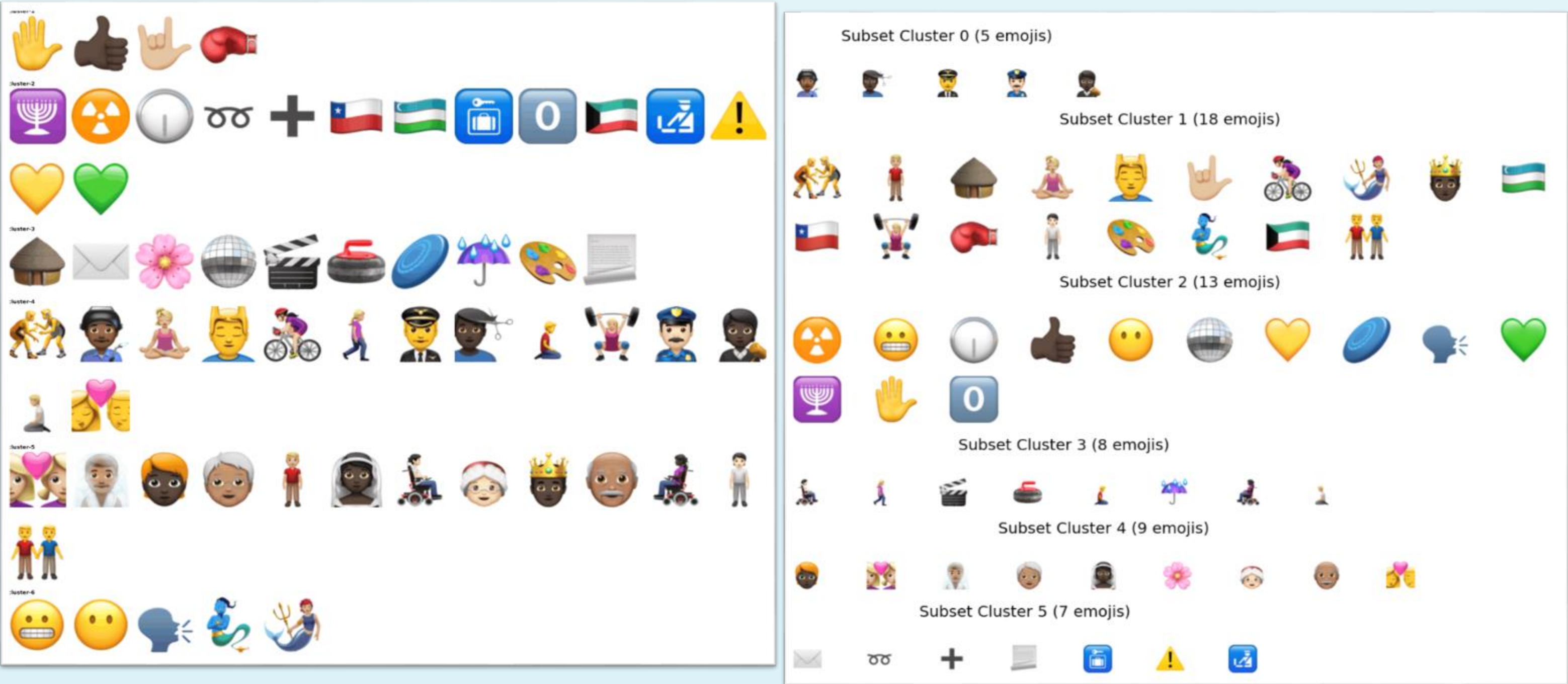
- Step 1** — Vision Extraction: EfficientNetB3 converted each emoji image into a 1,536-dimensional feature vector.
- Step 2** — Text Extraction: SentenceTransformers (all-MiniLM-L6-v2) converted each emoji name into a 384-dimensional semantic vector.
- Step 3** — Clustering: K-Means clustering was applied to both feature sets. The Silhouette Method was used to determine the optimal k (48 for vision, 43 for text).
- Step 4** — Human Card Sort: A custom web interface presented 60 random emojis to participants for manual grouping.
- Step 5** — Comparison: Pairwise F1 scores were used to measure agreement between human groupings and each AI model.

Conclusion

Humans categorize emojis based on semantic meaning rather than visual traits. Even though emojis are images, people think about them based on what they mean, not what they look like. This is why the NLP model, which understands meaning, did a better job of grouping emojis the way humans do than the vision model, which only sees pixels.

Results

Visual Features (Left) vs. Semantic Meaning (Right)

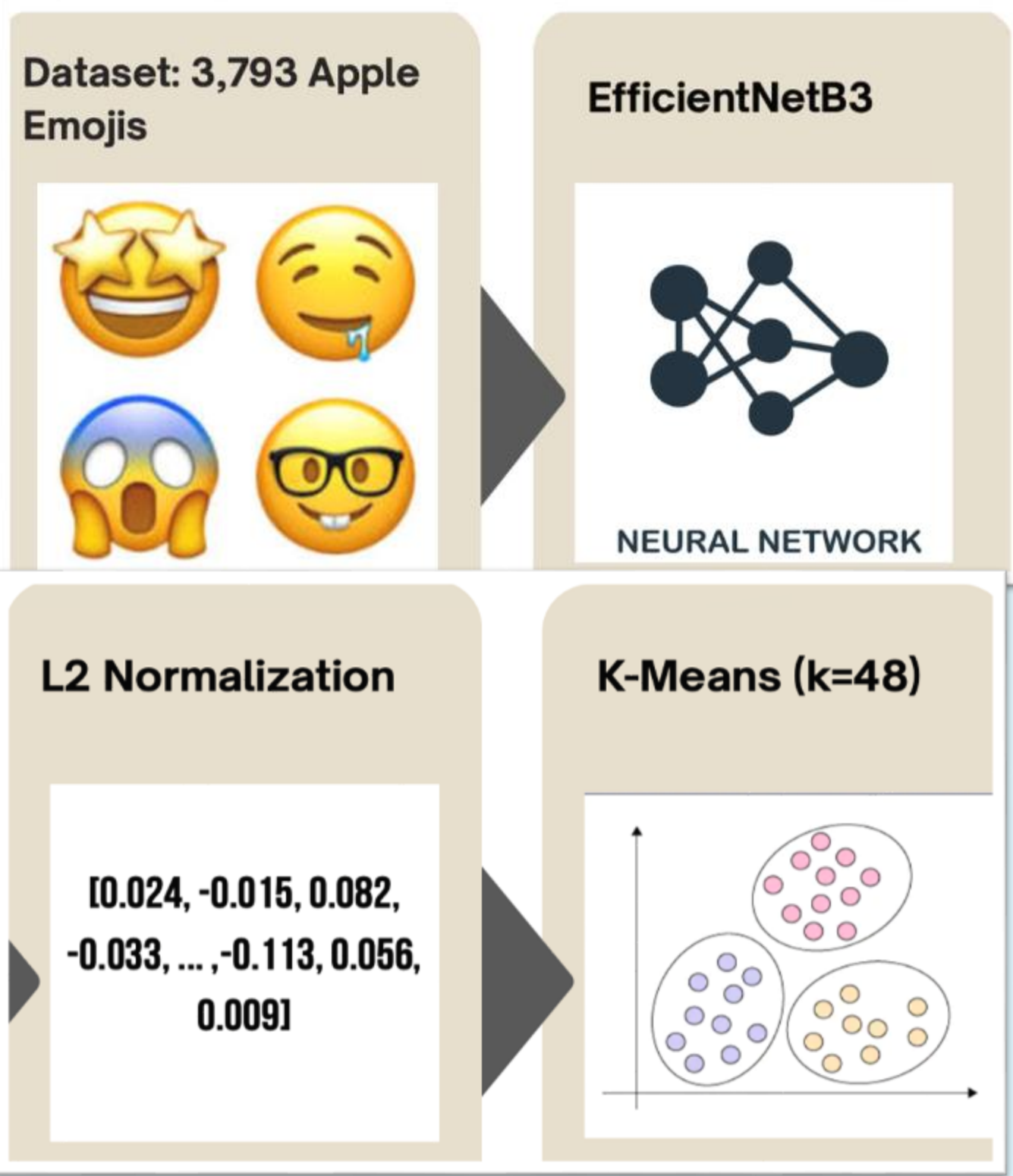


AI vs Human Clustering Comparison	
Participant: Person 1	
Precision : 0.558	
Recall : 0.22	
F1 Score : 0.315	
(TP=182, FP=144, FN=647)	
Participant: Person 2	
Precision : 0.181	
Recall : 0.316	
F1 Score : 0.23	
(TP=59, FP=267, FN=128)	
Participant: Person 3	
Precision : 0.279	
Recall : 0.283	
F1 Score : 0.281	
(TP=91, FP=235, FN=230)	
Average F1: 0.275	

Text-Based AI vs Human Clustering	
Participant: Person 1	
Precision : 0.624	
Recall : 0.682	
F1 Score : 0.651	
Participant: Person 2	
Precision : 0.109	
Recall : 0.529	
F1 Score : 0.181	
Participant: Person 3	
Precision : 0.283	
Recall : 0.798	
F1 Score : 0.417	
Text-based Average F1 : 0.416	
Vision-based Average F1: 0.275	
Improvement : +0.141	

Technical Stack

- Language: Python 3.11
- Environment: Jupyter Notebook
- Models:
- Vision: **EfficientNetB3**
- NLP: SentenceTransformers (**all-MiniLM-L6-v2**)
- Algorithms: K-Means Clustering, Principal Component Analysis (PCA) for dimensionality reduction.
- Evaluation: Silhouette Method and Pairwise F1 Score



Recommendations

- Multi-modal Fusion:** Develop a hybrid model that processes both visual pixels and semantic text simultaneously to capture physical and conceptual traits together.
- Hierarchical Clustering:** Move beyond K-Means to allow for sub-clusters, such as a Food group that branches into Fruit and Vegetables, better reflecting the way humans naturally organize information.

Acknowledgements

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